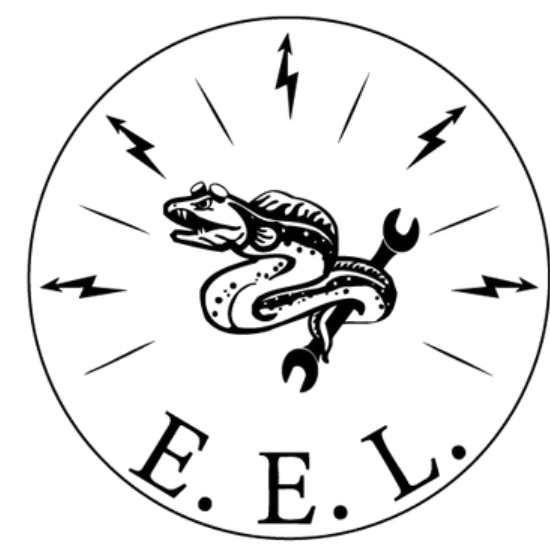




Reliable 3D Reconstruction with Radiance Fields

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Research Questions

1. In what ways do radiance field-based methods (NeRF, 3DGS) perform better than photogrammetry on a given dataset?
2. What aspects of NeRFs and Gaussian Splats can we combine with photogrammetry to create a high-fidelity portable experience?

Background

Photogrammetry - the practice of combining photographs of an object into a 3D textured mesh (Structure from Motion)

Data collected before

- >500 GB photos of Swayambhunath Mahachaitya temple complex
 - Organized into
- Laser-scanned point cloud
- Aerial photography



Fig. 1: Photogrammetry model manually simplified for web browser

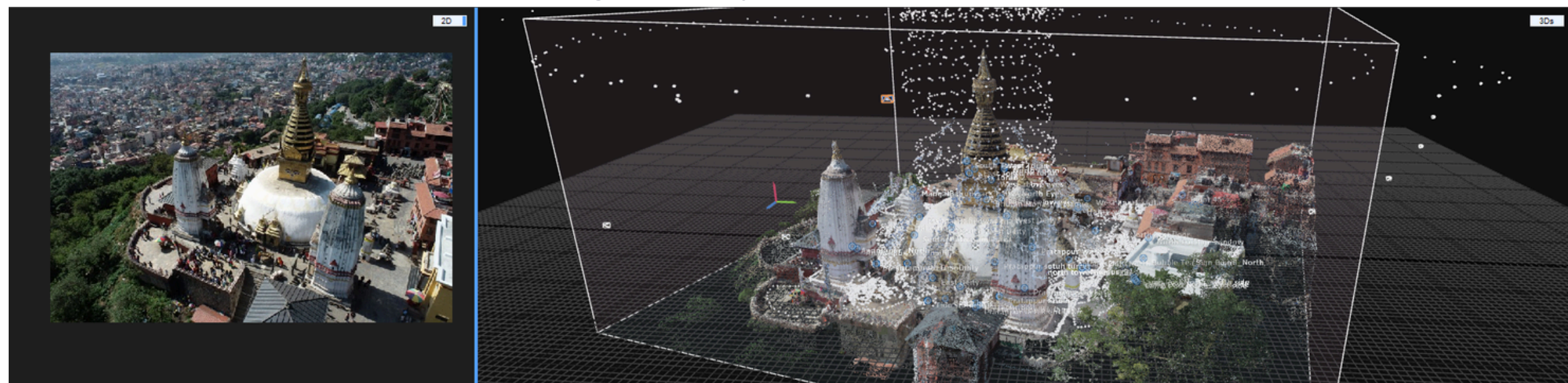
Photogrammetry Data

The Swayambhu Stories project originally used photogrammetry to reconstruct the cultural site as a 3D walkthrough tour. Some of the benefits and drawbacks of this approach include:

- + Better-documented, stable
- + Creates production asset
- Processing takes days
- Poor high-poly UV mapping and topology
- Unnecessarily large model sizes (15GB)

For this project, alignments and original images were selected as scene components and exported in COLMAP format as separate datasets of about 2000 images each. One combined dataset (14,000+ images) was formed for the whole scene, which could not be used due to memory constraints.

Fig. 2: RealityScan model view



Radiance Fields

A novel way to represent a 3D scene as a continuous volumetric function from which new images can be synthesized for any viewpoint.

$$\text{RF}(x,y,z,\theta,\phi) = \{\text{density}, \text{color}\} = \{\sigma(x,y,z), \mathbf{c}(x,y,z,\theta,\phi)\}$$

Neural Radiance Field (NeRF) (Mildenhall et al. 2020)

Infer the ideal parameters for a radiance field that resembles the given scene (from all given viewpoints)

3D Gaussian Splatting (3DGS) (Kopanas et al. 2023)

Places 3D “Gaussians” on a pre-defined point cloud of the scene, scaling and optimizing to minimize loss against input image. Alternative representation of RF

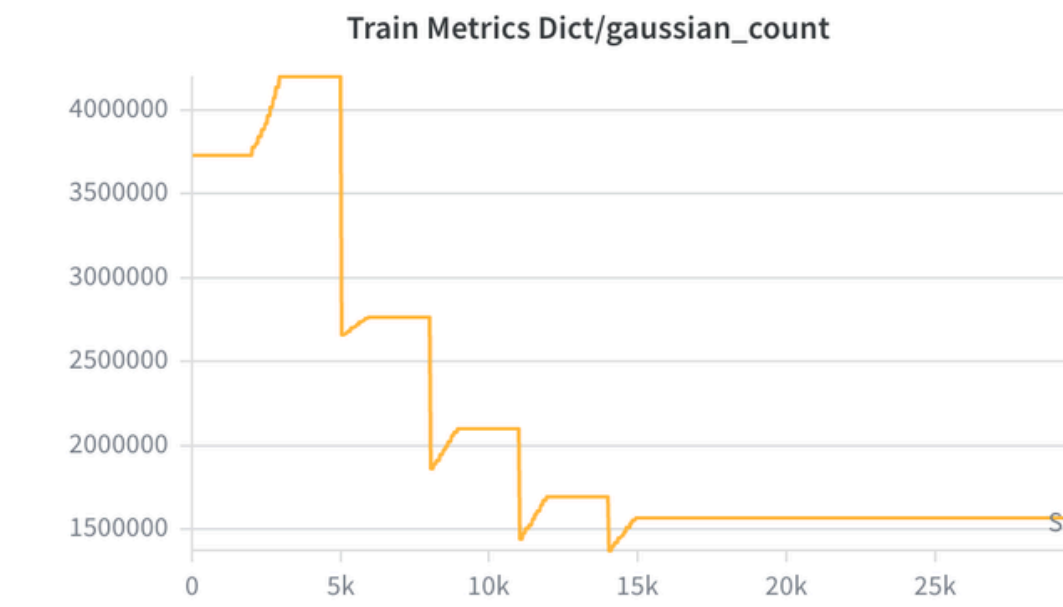


Fig. 3: Gaussian count plot; note densification, optimization, pruning stages

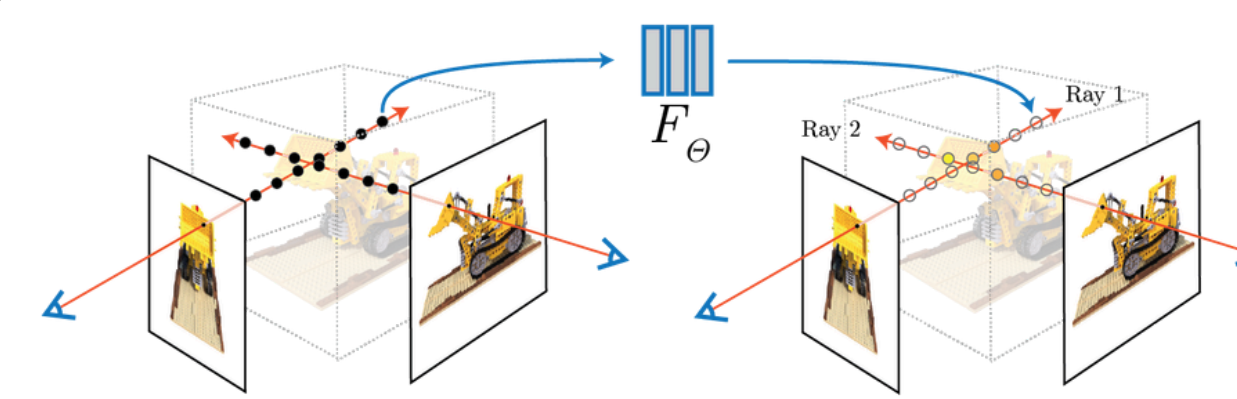


Fig. 4: NeRF (Mildenhall 2020)

NeRF

- + Scalable resolution for viewing
- + Provides diffuse map for baking
- + Can work alongside NeuS (Neural SDF, surface reconstruction)
- + Can use multi-GPU acceleration
- Repeated model query causes high overhead when generating real-time novel views (low FPS)
- Difficult and heavy to export
- Unnatural look from farther-away viewpoints
- + Scalable training parameters (num. Gaussians, growth/pruning rate)
- + Easier to render real-time
- + Trains significantly faster
- + Has some explicit 3D structure
- + Integrates with point clouds to avoid worse floaters
- + Easy to export
- Harder to optimize for multi-GPU
- Treats sky like a surface
- Can hit OOM mid-training (MCMC)

$$f_i(p) = \sigma(\alpha_i) \exp\left(-\frac{1}{2}(p - \mu_i)^T \Sigma_i^{-1} (p - \mu_i)\right)$$

Fig. 5: Gaussian (Yurkova, 2023)

> 3DGS <



Fig. 6: NeRF (Nerfacto) method early evaluation image



Fig. 7: 3DGS Markov Chain Monte Carlo (MCMC) strategy

Related Papers

- **Sparse Voxels Rasterization: Real-time High-fidelity Radiance Field Rendering** (Sun et al., Dec. 2025):
 - Allows real-time rendering of radiance fields without the moving artifacts of Gaussian splats or the inference overhead of NeRFs
- **NeRF-XL: Scaling NeRFs with Multiple GPUs** (Li et al., Apr. 2024)
 - Method created for use with larger datasets like ours
- **NeRFMeshing: Distilling [NeRFs] into Geometrically-Accurate 3D Meshes** (Rakotosaona et al., Mar. 2023)
 - Works with our method to create usable 3D meshes for retopology and optimization

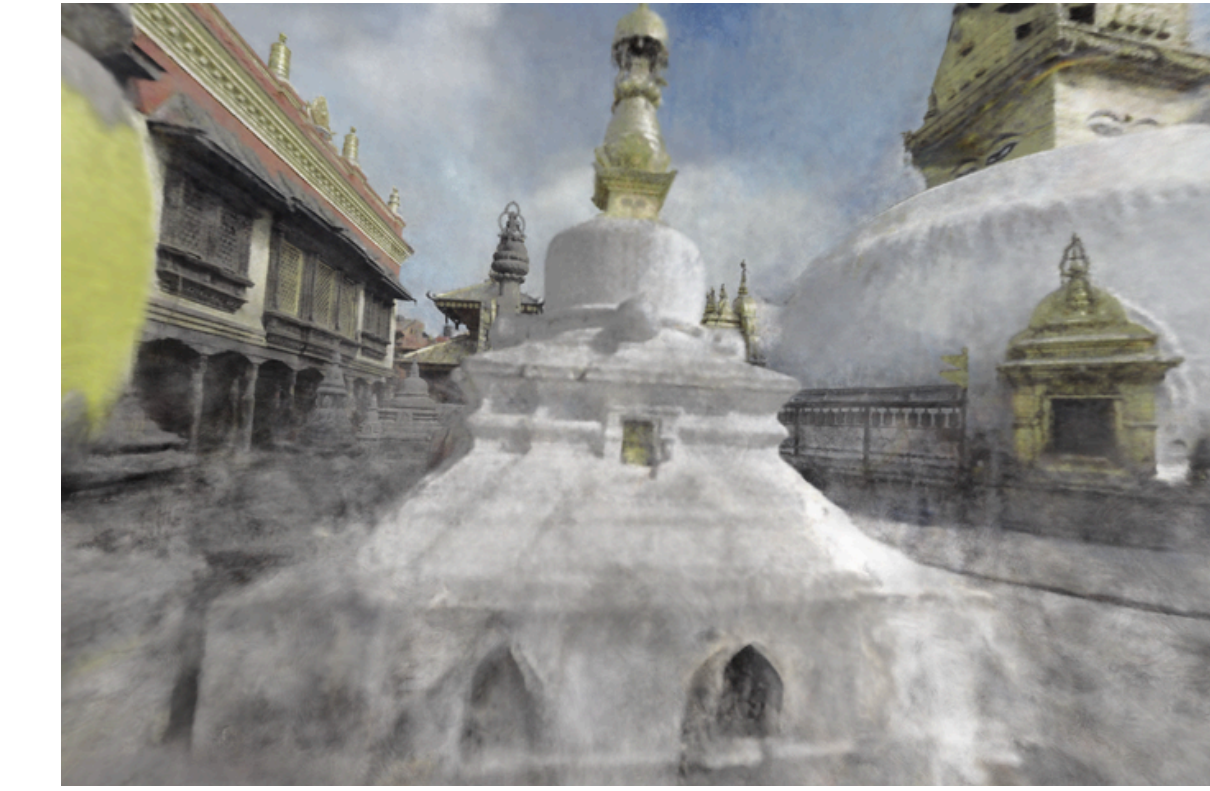
Results

As most scenes performed similarly and the combined scene resulted in OOM crashes, the following results are for one component scene.



Figs. 8 & 9: 3DGS, original view (left) and novel view (right)

Method	PSNR	SSIM
splatfacto-mcmc	13.6	<0.5
splatfacto-big	22.1	0.73
nerfacto-big	18.5	0.59
nerfacto-big (2 GPUs)	19.5	0.67



Figs. 9 & 10: Evaluation metrics (left), NeRF ~60% trained (right)

Future Considerations

Setting up the training environment was the most demanding task, as numerous build tools needed to be integrated. Data export from RealityScan into the Nerfstudio environment was also not adequately documented by the community and presented many ad-hoc problems.

In the future, we can look at using Gaussian Splats as a demonstration medium for VR and lower-compute-capability devices. Gaussian growth rate needs to be adjusted depending on the volume of data to avoid out-of-memory errors during training, but can often lead to divergent or poorly trained outcomes when compensating too heavily. NeRFs require too much power for a larger scene, but can be used to bake textures, create pre-rendered flyover videos and similar media. Additionally, NeRF and NeuS can be tested and combined in the future as a full reconstruction suite to gauge the reliability of neural network-based methods against traditional photogrammetry.

References

- Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian splatting for real-time radiance field rendering. ACM Transactions on Graphics (SIGGRAPH).
- Li, Z., et al. (2024). NeRF-XL: Scaling NeRFs with multiple GPUs.
- Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. In European Conference on Computer Vision (ECCV).
- Rakotosaona, M.-J., et al. (2023). NeRFMesh: Distilling NeRFs into geometrically-accurate 3D meshes. Sun, X., et al. (2025). Sparse voxels rasterization: Real-time high-fidelity radiance field rendering.
- Towards Data Science. (2023). A comprehensive overview of Gaussian splatting. <https://towardsdatascience.com/a-comprehensive-overview-of-gaussian-splatting-e7d570081362>